FarmFi

Decentralized Agricultural Micro-Finance Protocol on Botchain

Whitepaper v1.0 | Botchain Agriculture Series | November 2026

1. Executive Summary

FarmFi is a decentralized agricultural micro-finance protocol deployed on Botchain. It connects smallholder farmers seeking seasonal production loans with capital providers (lenders) looking for yield, facilitated entirely by transparent smart contracts. Loan rates are dynamically set by an on-chain credit scoring engine powered by repayment history.

An estimated 65% of smallholder farmers in sub-Saharan Africa lack access to formal credit. FarmFi eliminates the need for traditional banks by replacing credit bureaus and loan officers with immutable on-chain data.

2. Problem Statement

2.1 Financial Exclusion of Farmers

Rural farmers are routinely denied bank loans due to lack of credit history, collateral, or documentation. They resort to predatory informal lenders charging 50-200% interest, eroding farming profits and deepening poverty cycles.

2.2 Idle Lender Capital

DeFi participants on Botchain lack targeted yield opportunities in the real-economy agricultural sector. FarmFi creates a productive, impact-positive use case for idle BOT capital.

3. Protocol Architecture

3.1 Lending Pool

Lenders deposit BOT into a shared liquidity pool. Interest earned from farmer repayments is distributed to lenders proportionally via an accRewardPerShare accumulator pattern, ensuring gas-efficient yield distribution.

3.2 Credit Scoring Engine

Every farmer starts with a credit score of 500 (out of 1000). On-time repayments increase the score by 20 points (capped at 1000). Defaults reduce it by 100 points. The score directly determines the interest rate applied to the next loan.

Credit Score	Interest Rate	Classification
800 – 1000	6% p.a.	Excellent
600 – 799	8% p.a.	Good
400 – 599	10% p.a.	Average
0 – 399	12% p.a.	High Risk

3.3 Collateral System

Farmers declare a crop or asset as collateral with a minimum 150% collateral-to-loan ratio. Collateral is described on-chain (e.g., "500kg tomato harvest") and enforced socially via verifier assessors, with physical liquidation triggered by the platform in default cases.

3.4 Seasonal Loan Types

- Planting Loans: fund seeds, fertiliser, and land preparation
- Growing Loans: fund irrigation, pest control during growing season
- Harvest Loans: fund harvest labor and packaging
- Post-Harvest Loans: fund storage and transport to market

4. Milestones & Activities

Milestone	Activity	Timeline	KPI
M1 – Testnet Deploy	Deploy FarmFi.sol; full test suite for deposit, borrow, repay	Month 1	100% test coverage
M2 – Lender Onboarding	Bootstrap lending pool with 50 BOT from founding lenders	Month 2	50 BOT pool minimum
M3 – Farmer Pilots	Onboard 20 pilot farmers; first 5 loans disbursed	Months 2–3	5 active loans
M4 – Credit History Build	Track 3-month repayment cycle; credit scores populated	Months 3–5	Score data for 20 farmers
M5 – dApp Launch	Launch FarmFi frontend; wallet integration and loan dashboard	Month 3	Live dApp
M6 – Verifier Network	Recruit 5 field verifiers for collateral assessment	Month 4	5 certified verifiers
M7 – Mainnet Expansion	Scale to 200 farmers; target 500 BOT pool	Month 6	500 BOT TVL

5. Risk Management

- Default Reserve: 10% of premiums held in reserve to absorb defaults
- Max 2 active loans per farmer to limit concentration risk
- Dynamic interest rates disincentivise risky borrowing behaviour
- Owner-controlled markDefault function triggers at loan maturity
- Credit score penalty creates strong repayment incentives

6. Economic Model

FarmFi is a non-profit protocol. No platform fee is extracted. 100% of interest paid by farmers flows to the lending pool, distributed to depositors. The protocol sustains itself through the natural incentive alignment between farmers (cheap credit) and lenders (real-economy yield).

7. Conclusion

FarmFi brings DeFi credit infrastructure to the last mile of the agricultural economy. By deploying on Botchain, FarmFi benefits from low transaction costs and fast finality, making micro-loans as small as 0.1 BOT economically viable. With on-chain credit scoring, every farmer who repays a loan builds a permanent, portable financial identity that will unlock better rates and larger loans over time.